Reg. No. | | | | | | | | | | | [12EE72]

SEVENTH SEMESTER B.E EXAMINATION, DECEMBER–2015

UTILIZATION OF ELECTRICAL POWER

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full questions.

1.	**a.**	State the advantages of electrical heating.	**04**
	b.	With a neat diagram explain the working of direct arc furnace.	**06**
	c.	Write the application of dielectric heating.	**04**
2.	**a.**	With neat diagram explain any two types of resistance welding.	**10**
	b.	Discuss briefly equipment's used for electric arc welding.	**04**
3.	**a.**	Calculate the ampere hours required to deposit a coating of silver 0.05mm thick on a sphere of 5 cm radius assume electrochemical equivalent of silver is 0.001118 and density of silver to be 10.5 gm/cm^3.	**06**
	b.	With a neat sketch explain the refining process of copper.	**08**
4.	**a.**	State and explain laws of illumination.	**04**
	b.	A lamp of 500w having mscp of 1000 is suspended 2.7 m above the working plane calculate i. Illumination directly below the lamp at the working plane ii. Illumination at a point 2.5m away on the horizontal plane from vertically below the lamp	**06**
	c.	Compare incandescent lamp and fluorescent lamp.	**04**
5.	**a.**	Discuss briefly different systems of traction.	**10**
	b.	Compare AC and DC traction system.	**04**
6.	**a.**	The speed time curve of a train consist of: i. Uniform acceleration of 6km/h/s for 25 sec ii. Uniform deceleration of 6km/h/s to stop the train iii. A stop of 5 minutes Find the distance between the stations, the average and schedule speeds.	**08**
	b.	What are the factors affecting specific energy consumption?	**06**
7.	**a.**	Write short notes on: i. Regenerative braking ii. Energy saving by series parallel control	**07** **07**
8.	**a.**	With a block diagram explain the working principle of hybrid vehicles.	**08**
	b.	With a neat diagram explain the characteristics of traction motors.	**06**

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